Rajeev Jain

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Principal Specialist, Research Software Engineering | AI Systems and Verification | HPC | Scientific Computing

At Argonne National Laboratory **since 2009** | **2× R&D 100 Awards** (CANDLE 2023, FLASH-X 2022) | Lead developer, **UXarray** (DOE labs & NCAR)

Professional Profile

Principal research software engineer, at Argonne since 2009, delivering production systems at the intersection of AI infrastructure, scientific computing, and HPC. Led UXarray from zero to a Python library used at DOE labs and NCAR, including an MCP server that lets AI agents inspect, analyze, and visualize production Earth-system meshes on HPC clusters via Globus Compute no SSH, no scripts, full provenance. Pairs that with a verification practice: measuring what AI tooling and machine-assisted code actually do rather than accepting the reported figure, which has surfaced silent numerical drift, hidden heap allocation in a faithful port, and vendor throughput numbers that are unreachable by construction. Ran 10,000+ ML training experiments using Swift/T across Summit, Theta, and Cori for cancer drug response research; applied noise injection and counterfactual analysis to identify genes driving tumor classification decisions. Reduced FLASH-X checkpoint overhead 40–70% on Summit via async HDF5 and lossy compression. Generated a 101M-element MONJU reactor mesh on 712 processors in about 7 minutes a mesh the serial path could not build at all. Two R&D 100 Awards (2022, 2023). Work spans Python platforms and MCP server development, C++ and Fortran HPC codebases, exascale GPU porting (Intel XPU / CUDA), and cross-institution program coordination.

Education

University of Chicago — 2020
M.S. in Computer Science

Arizona State University — 2009
M.S. in Structural Engineering
Advisors: Dr. S.D. Rajan (ASU), Dr. Ashok Belegundu (PSU)
Graduate Fellowship Recipient (2007–2009)

Indian Institute of Technology (ISM) Dhanbad — 2006
B.Tech. in Mechanical Engineering

Professional Experience

Argonne National Laboratory — **2009 – Present**
Principal Specialist, Research Software Engineering — *Mathematics and Computer Science Division, Chicago, IL*

- Principal software engineer and program lead across six multi-year DOE research software programs spanning climate analytics, scientific AI, simulation infrastructure, and exascale HPC porting from initial architecture through production release, CI/CD, and cross-institution coordination.

University of Chicago — **2023 – Present**
Staff At-Large — *Consortium for Advanced Science and Engineering (CASE), Chicago, IL*

- Joint appointment supporting cancer pharmacogenomics and Earth science collaborations.
- Mentor students and research staff on software design, experimentation, and reproducibility.

Arizona State University — **2007 – 2009**
Research and Teaching Assistant — *Structural and Computational Mechanics Lab, Tempe, AZ*

- Conducted FEM-based research in blast mitigation and supported structural analysis coursework.

Earlier Industry Experience — **2005 – 2007**
Project Engineer, Wipro Technologies; Engineering Intern, Tata Motors — *India*

Research Software Leadership and Projects

UXarray **2022 – Present**
Lead Developer *Open-source climate analysis platform*

- Lead developer for a Python library enabling analysis of native unstructured climate and weather grids, used by NCAR, DOE labs, and universities.
- Designed the core API, conservative zonal averaging via exact spherical polygon intersection, vector calculus operators (gradient, divergence, curl), YAC remapping backend, multi-grid support, geometry operators, grid I/O for ESMF / MPAS / SCRIP / HEALPix, automated tests, CI/CD, and release workflows.
- Built the UXarray MCP server: 20+ typed tools allowing AI agents to inspect, subset, visualize, and analyze production Earth-system meshes locally or on HPC clusters (Argonne Improv, UCAR Derecho) via Globus Compute, with structured provenance on every operation. Presented at SciFoundationModels 2026 (SciFM26).
- Tutorials and presentations at SC24, AMS 2024/2025/2026, SciPy 2023, EGU 2024/2025/2026, AGU, and ESDS.

AI Code and Benchmark Verification **2025 – Present**
Independent *Measurement practice for AI tooling and machine-assisted code*

- Published a reproducible benchmark harness for local LLM inference (scripts, raw data, and protocol, MIT-licensed) covering four model architectures on one machine. Established that bytes read per generated token must be parsed from the GGUF tensor table rather than taken from file size, which overstates by 64% on models using gathered per-layer embeddings, and that the memory-bandwidth denominator must be measured (340 GB/s) rather than read from the 400 GB/s datasheet pin rate.
- Result: all four architectures extract 36–41% of memory bandwidth, so an 8× spread in tokens per second is explained entirely by a 6.8× spread in bytes per token — throughput is a property of model size and runtime efficiency, not architecture. Protocol records background load as a covariate rather than gating on it, and reports the mean of two order-reversed passes with the pass-to-pass spread visible.
- Correctness work in numerical Python: traced 0.68 m of drift on a production Earth mesh to catastrophic cancellation inside a spherical cross product — a difference of nearly equal products, not a running sum — and proposed error-free-transformation arithmetic (`two_prod`) for the kernel rather than compensated summation, which does not address that failure mode (merged upstream, UXarray PR #1513); recovered a 1.67× speedup in a Numba hot path by eliminating heap allocation that a faithful C++-to-Python translation had silently introduced.
- Wrote the case against general-purpose chat rollout as an AI strategy, arguing for typed, scoped, provenance-tracked tools as the architecture that actually returns value.

SFNO Climate Emulator on Aurora **2024 – Present**
Software Engineer *AI climate emulation on exascale Intel GPUs*

- Ported the SFNO climate emulator to Intel XPU on Aurora: PMIX/PALS launcher mapping, XPU/CUDA device branching, device-aware AMP (gradient scaling on CUDA, bf16 on XPU), and first stable DDP baseline on Aurora.
- Established smoke-test and queue-aware bring-up scripts; separated launcher, backend, and device-placement failure modes from sharding bugs.

CANDLE / IMPROVE **2017 – 2025**
Core Contributor *Cancer AI infrastructure and benchmarking*

- Built the CANDLE/Supervisor HPO workflow using Swift/T, mlrMBO (Bayesian/GP), DEAP (genetic algorithms), and Hyperopt/TPE; ran 10,000+ model training experiments across Summit, Theta, and Cori at up to 1,000 concurrent jobs.
- Applied noise injection and counterfactual analysis to the NT3 tumor classification benchmark; identified PLOD2, RGS5, TP53I13, and related genes as decision-boundary drivers confirming the model used biologically meaningful signal.
- Led the IMPROVE benchmark framework: `improvelib` Python package with standardized loaders, canonical train/test splits, and evaluation metrics; GitHub Actions CI/CD covering unit tests, end-to-end Docker pipelines, and parallel cross-study analysis (CSA) via Parsl.
- Designed and implemented UNO, a dual-branch neural network (separate drug and cell-line embeddings) that participates in the IMPROVE CSA benchmark; results published in *Briefings in Bioinformatics* (2026).
- Supported a 15+ researcher multi-lab collaboration. R&D 100 Award, 2023.

FLASH-X **2019 – 2024**
I/O and Compression Lead *Multiphysics simulation software*

- Implemented asynchronous HDF5 I/O with Argobots and integrated SZ3 / ZFP compression for checkpoint and restart workflows.
- Reduced checkpoint overhead by 40-70% on Summit and achieved storage savings above 50%.

- Enabled cross-checkpoint restart between AMReX and Paramesh solvers. Work contributed to an R&D 100 Award in 2022.

MeshKit **2009 – 2016**
Principal Investigator / Software Lead *DOE NEAMS reactor mesh generation*

- Principal investigator for MeshKit within DOE NEAMS, leading open-source C++ toolkit for automated nuclear reactor core mesh generation.
- Developed parallel CoreGen workflow: 712 processors, 101 million hexahedral elements, 14 GB MONJU reactor mesh in about 7 minutes serial path could not run at all.
- Built AssyGen/CoreGen pipeline encoding reactor lattice hierarchy; supported PostBL boundary-layer generation and multi-format mesh workflows.

Urban ECP **2016 – 2019**
Software Lead *DOE urban microclimate and CFD workflows*

- Led seed-funded urban exascale proposal with Charlie Catlett (Argonne / University of Chicago); developed a three-layer coupled simulation workflow: WRF/HRRR mesoscale weather → EnergyPlus building energy → Nek5000 wall-resolved LES for downtown Chicago.
- Built the full 3D urban geometry model for the CFD domain, including architecturally complex structures such as Lake Point Tower.
- Published in *Journal of Building Performance Simulation* (2020); presented at AGU 2018/2019 and International Conference on Urban Climate 2018.

Funding and Research Programs

- **DOE SEATS** (active): Simplifying ESM Analysis Through Standards.
- **NSF Raijin** (active): collaborative research in climate model analysis.
- **DOE ECP CANDLE** (2017 – 2023): core contributor on cancer AI infrastructure.
- **DOE NEAMS** (2009 – 2016): principal investigator for MeshKit and reactor mesh generation workflows.
- **Urban ECP** (2016 – 2019): software lead and coordinator for urban microclimate and CFD workflows across multiple labs.

Awards & Recognition

- R&D 100 Award — CANDLE, Cancer Distributed Learning Environment (2023).
- R&D 100 Award — FLASH-X, multiphysics simulation software (2022).
- DOE SBIR Phase I & II — RGG (Reactor Geometry & mesh Generator) tool, awarded to Kitware Inc. (2014 – 2017).
- ATPESC Scholar, Argonne Training Program on Extreme-Scale Computing (2015).
- Best Paper Award, International Meshing Roundtable (2010).
- University Graduate Fellowship, Arizona State University (2007 – 2009).
- First Prize, Low-Budget Car Design Contest — Minda Ltd. & IIT Kharagpur (2004).

Peer-Reviewed Journal Articles

2026 Chen, H., Ullrich, P. A., Panetta, J., Marsico, D., Hanke, M., **Jain, R.**, Zhang, C., et al. "Accurate and Robust Geometric Algorithms for Regridding on the Sphere." *EGUsphere* (preprint).

2026 Partin, A., Vasanthakumari, P., Narykov, O., Wilke, A., Koussa, N., Jones, S., Zhu, Y., **Jain, R.**, Overbeek, J. C., Fernando, G. D., Sanchez-Villalobos, C., Garcia-Cardona, C., Mohd-Yusof, J., Chia, N., Wozniak, J. M., Ghosh, S., Pal, R., Brettin, T. S., Weil, M. R., and Stevens, R. L. "Benchmarking community drug response prediction models: datasets, models, tools, and metrics for cross-dataset generalization analysis." *Briefings in Bioinformatics* 27(1), bbaf667.

2022 Dubey, A., Weide, K., O'Neal, J., Dhruv, A., Couch, S., Harris, J. A., Klosterman, T., **Jain, R.**, Rudi, J., Messer, B., Pajkos, M., Carlson, J., Chu, R., Wahib, M., Chawdhary, S., Ricker, P. M., Lee, D., Antypas, K., Riley, K. M., Daley, C., Ganapathy, M., Timmes, F. X., Townsley, D. M., Vanella, M., Bachan, J., Rich, P. M., Kumar, S., Endeve, E., Hix, W. R., Mezzacappa, A., and Papatheodore, T. "Flash-X: A multiphysics simulation software instrument." *SoftwareX*.

2022 Harris, J. A., Chu, R., Couch, S. M., Dubey, A., Endeve, E., Georgiadou, A., **Jain, R.**, Kasen, D., Laiu, M. P., Messer, O. E. B., O'Neal, J., Sandoval, M. A., and Weide, K. "Exascale models of stellar explosions: quintessential multi-physics simulation." *International Journal of High Performance Computing Applications*.

2020 **Jain, R.**, Luo, X., Sever, G., Hong, T., and Catlett, C. "Representation and evolution of urban weather boundary conditions in downtown Chicago." *Journal of Building Performance Simulation.*

2018 Wozniak, J. M., **Jain, R.**, Balaprakash, P., Ozik, J., Collier, N. T., Bauer, J., Xia, F., Brettin, T., Stevens, R., Mohd-Yusof, J., Garcia-Cardona, C., Van Essen, B., and Baughman, M. "CANDLE/Supervisor: A workflow framework for machine learning applied to cancer research." *BMC Bioinformatics.*

2014 Mahadevan, V. S., Merzari, E., Tautges, T., **Jain, R.**, Obabko, A., Smith, M., and Fischer, P. "High-resolution coupled physics solvers for analysing fine-scale nuclear reactor design problems." *Philosophical Transactions of the Royal Society A.*

2012 Tautges, T. J. and **Jain, R.** "Creating geometry and mesh models for nuclear reactor core geometries using a lattice hierarchy-based approach." *Engineering with Computers.*

2008 Argod, V., Belegundu, A. D., Aziz, A., Agrawala, V., **Jain, R.**, and Rajan, S. D. "MPI-enabled shape optimization of panels subjected to air blast loading." *International Journal for Simulation and Multidisciplinary Design Optimization.*

Peer-Reviewed Conference and Workshop Proceedings

2026 **Jain, R.** and Jacob, R. "Beyond Tool Execution: Evaluating Scientific MCP Interfaces with UXarray." *1st Workshop on Agentic AI for Large-scale Science (AGENT4SC) at IEEE eScience 2026*, Naples, Italy. Accepted; to be presented. Artifacts and data: doi:10.5281/zenodo.21463232.

2025 Gwinn, J., Wozniak, J. M., **Jain, R.**, Zhu, Y., Partin, A., Brettin, T., and Stevens, R. "A Workflow for Error Analysis for Drug Response Prediction via Statistical Standardization and Distribution Analysis." *Workshop on Workflows in Support of Large-Scale Science (WORKS).*

2024 **Jain, R.**, Tang, H., Dhruv, A., and Byna, S. "Enabling Data Reduction for Flash-X Simulations." *SC24 Workshops: DRBSD-10.*

2024 **Jain, R.**, Wozniak, J. M., Partin, A., Wilke, A., Zhu, Y., Vasanthakumari, P., Narykov, O., Overbeek, J., Weaver, R., Wang, C., Liu, Y., Weil, M. R., Brettin, T., and Stevens, R. "Cross-HPO: Optimizing Neural Networks for Cancer Drug Response Using Hyperparameter Tuning on Multiple Pharmacogenomic Datasets." *Computational Approaches for Cancer Workshop at SC24.*

2023 Wozniak, J. M., **Jain, R.**, Wilke, A., Weaver, R., Partin, A., Brettin, T., and Stevens, R. "An Automation Framework for Comparison of Cancer Response Models Across Configurations." *IEEE e-Science.*

2023 Dhruv, A., **Jain, R.**, O'Neal, J., Weide, K., and Dubey, A. "Framework and Methodology for Verification of a Complex Scientific Simulation Software, Flash-X." *Congress in Computer Science, Computer Engineering, and Applied Computing.*

2022 **Jain, R.**, Tang, H., Dhruv, A., Harris, J. A., and Byna, S. "Accelerating Flash-X Simulations with Asynchronous I/O." *PDSW.*

2021 **Jain, R.**, Shah, A., Mohd-Yusof, J., Wozniak, J. M., Brettin, T., Xia, F., and Stevens, R. "Probing Decision Boundaries in Cancer Data Using Noise Injection and Counterfactual Analysis." *CAFCW at SC21.*

2021 **Jain, R.**, Weide, K., Chawdhary, S., and Klostermann, T. "Checkpoint / Restart for Lagrangian Particle Mesh with AMR in Community Code FLASH-X." *SuperCheck.*

2015 **Jain, R.** and Tautges, T. J. "NEAMS MeshKit: Nuclear Reactor Mesh Generation Solutions." *International Congress on Advances in Nuclear Power Plants.*

2015 **Jain, R.**, Mahadevan, V., and O'Bara, R. "Simplifying Workflow for Reactor Assembly and Full-Core Modeling." *Mathematics and Computation (MC2015).*

2014 **Jain, R.** and Tautges, T. J. "Generating Unstructured Nuclear Reactor Core Meshes in Parallel." *Procedia Engineering.*

2013 **Jain, R.** and Tautges, T. J. "PostBL: Post-Mesh Boundary Layer Mesh Generation Tool." *International Meshing Roundtable.*

2012 **Jain, R.** and Tautges, T. J. "RGG: Reactor Geometry (and Mesh) Generator." *International Congress on Advances in Nuclear Power Plants.*

2012 Mohanty, S., **Jain, R.**, Majumdar, S., Tautges, T. J., and Srinivasan, M. "Coupled Field Structural Analysis of HTGR Fuel Brick Using Abaqus." *International Congress on Advances in Nuclear Power Plants.*

2010 Tautges, T. J. and **Jain, R.** "Creating Geometry and Mesh Models for Nuclear Reactor Core Geometries Using a Lattice Hierarchy-Based Approach." *19th International Meshing Roundtable.* Best Paper Award.

Conference Abstracts, Tutorials, and Presentations

2027 **Jain, R.** "Accurate Conservative Zonal Averaging for Unstructured Earth-System Grids: A Python Implementation of Robust Spherical Geometry Algorithms." *107th AMS Annual Meeting.* Accepted.

2026 **Jain, R.**, Abdulla, D., Jacob, R., Clyne, J., Eroglu, O., Medeiros, B., Chen, H., and Ullrich, P. "A Model Context Protocol Server for Agentic Analysis of Unstructured Earth-System Meshes." Presented at *SciFoundationModels 2026 (SciFM26).*

2026 Eroglu, O., Chen, H., Chmielowiec, P., Clyne, J., Hannay, C., Jacob, R. L., **Jain, R.**, Medeiros, B., Ullrich, P., and Zarzycki, C. M. "Latest Developments in UXarray: A Python Package Supporting Kilometer-Scale Model Outputs in Native Unstructured Grids." *106th AMS Annual Meeting.*

2026 Clyne, J., Eroglu, O., Jacob, R., **Jain, R.**, Medeiros, B., Ullrich, P., and Zarzycki, C. "UXarray: A Python Package for the Analysis of Kilometer-Scale Atmosphere and Ocean Model Outputs." *EGU General Assembly (EGU26).*

2025 Eroglu, O., Chen, H., Chmielowiec, P., Hannay, C., Jacob, R. L., **Jain, R.**, Medeiros, B., Ullrich, P., and Zarzycki, C. M. "UXarray: Extending Xarray to Support the Analysis of Native, Kilometer-Scale Unstructured Grids in Python." *105th AMS Annual Meeting.*

2025 Clyne, J., Chen, H., Chmielowiec, P., Eroglu, O., Hannay, C., Jacob, R., **Jain, R.**, Medeiros, B., Ullrich, P., and Zarzycki, C. "UXarray: Extending Xarray for Enhanced Support of Unstructured Grids." *EGU General Assembly.*

2024 **Jain, R.** "Tutorial: UXarray for Analysis of Unstructured Climate Data." *SC24.*

2024 Chmielowiec, P., Chen, H., DeCiampa, C., Eroglu, O., Hannay, C., Jacob, R. L., **Jain, R.**, Loft, R., Medeiros, B., Sun, L., Ullrich, P., and Zarzycki, C. M. "UXarray: Extending Xarray with Support for Unstructured Grids." *104th AMS Annual Meeting.*

2024 Eroglu, O., Chen, H., Chmielowiec, P., Clyne, J., DeCiampa, C., Hannay, C., Jacob, R., **Jain, R.**, Loft, R., Medeiros, B., Sun, L., Ullrich, P., and Zarzycki, C. "UXarray: Extensions to Xarray to Support Unstructured Grids." *EGU General Assembly.*

2024 Clyne, J., Chen, H., Chmielowiec, P., Hannay, C., Jacob, R. L., **Jain, R.**, Medeiros, B., Ullrich, P. A., and Zarzycki, C. M. "UXarray: An Extension to Xarray Supporting Analysis and Visualization of Kilometer-Scale Model Outputs." *AGU Fall Meeting.*

2024 **Jain, R.** "UXarray Tutorial." *ESDS Annual Event.* Video.

2023 **Jain, R.** "UXarray for Unstructured Climate Data." *SciPy 2023.*

2023 **Jain, R.** "Data Reduction for FLASH-X Simulations." *HDF User Group 2023.*

2023 Clyne, J., Chen, H., Chmielowiec, P., DeCiampa, C., Eroglu, O., Hannay, C., Jacob, R. L., **Jain, R.**, Medeiros, B., Sun, L., Ullrich, P. A., and Zarzycki, C. M. "UXarray: A Community-Developed Tool for the Analysis and Visualization of Model Output on Unstructured Grids in Python." *AGU Fall Meeting.*

2023 Jacob, R. L., Ullrich, P. A., Zhang, C., **Jain, R.**, Chen, H., Po-Chedley, S., and Vo, T. "Creating New Standards to Simplify Earth System Model Analysis." *AGU Fall Meeting.*

2019 Sever, G., Obabko, A., Jacob, R., **Jain, R.**, and Catlett, C. "Simulations of Idealized and Real Urban Boundary Layer Environments Using NWP and CFD Models." *18th Conference on Mesoscale Processes.*

2019 Jacob, R. L., Sever, G., Obabko, A., **Jain, R.**, and Catlett, C. E. "CFD Simulations of Flow in Realistic Urban Geometries Initialized from Weather Models." *AGU Fall Meeting.*

2018 Obabko, A., **Jain, R.**, Jacob, R., Paoli, R., Sever, G., Grindeanu, I., and Fischer, P. "Nek5000 Wall-Resolved LES for Urban Geometries." *10th International Conference on Urban Climate / 14th Symposium on the Urban Environment.*

2018 Silva, M. P., Sharma, A., Budhathoki, M., **Jain, R.**, and Catlett, C. E. "Neighborhood Scale Heat Mitigation Strategies Using Array of Things Data in Chicago." *AGU Fall Meeting.*

2018 Sever, G., Jacob, R., and **Jain, R.** "Mesoscale to Microscale Weather Simulations over the Chicago Downtown Region." *10th International Conference on Urban Climate / 14th Symposium on the Urban Environment.*

Technical Reports, Manuals, and Other Publications

2020 Goldring, N., Bruhwiler, D., Nash, B., Wu, Z., Nagler, R., Carter, J., Lerch, J., Suthar, K., Den Hartog, P., **Jain, R.**, and Mahadevan, V. "Multiphysics Design and Optimization of Complex Vacuum Chambers (Phase II Final Technical Report)."

2017 **Jain, R.** "Reactor Geometry (&Mesh) Generator Users Manual."

2016 **Jain, R.**, Vanderzee, E., Grindeanu, I., and Mahadevan, V. "Mesh Generation and Algorithm Development for NEAMS."

2015 **Jain, R.**, Vanderzee, E., and Mahadevan, V. "Update on Development of Mesh Generation Algorithms in MeshKit." ANL/MCS-TM-355.

2015 **Jain, R.** and Mahadevan, V. "Documentation for MeshKit-Reactor Geometry (&mesh) Generator." ANL/MCS-TM-354.

2015 Mahadevan, V., Grindeanu, I. R., Ray, N., **Jain, R.**, and Wu, D. "SIGMA Release v1.2: Capabilities, Enhancements and Fixes." ANL/MCS-TM-357.

2015 Merzari, E., Shemon, E. R., Yu, Y., Thomas, J. W., Obabko, A., **Jain, R.**, Mahadevan, V., Solberg, J., Ferencz, R., and Whitesides, R. "Full Core Multiphysics Simulation with Offline Mesh Deformation." ANL/NE-15/42.

2015 Merzari, E., Shemon, E. R., Yu, Y. Q., Thomas, J. W., Obabko, A., **Jain, R.**, Mahadevan, V., Tautges, T., Solberg, J., Ferencz, R. M., and Whitesides, R. "Multi-physics Demonstration Problem with the SHARP Reactor Simulation Toolkit." ANL/NE-15/44.

2014 **Jain, R.** "Report on FY11 Extensions to MeshKit and RGG." ANL/MCS-TM-316.

2014 **Jain, R.** and Mahadevan, V. "2014 MeshKit Release." ANL/MCS-TM-344.

2014 Tautges, T. J. and **Jain, R.** "Mesh Copy/Move/Merge Tool for Reactor Simulation Applications." ANL/MCS-TM-343.

2013 Tautges, T. J., Fischer, P., Grindeanu, I., **Jain, R.**, Mahadevan, V., Obabko, A., Smith, M. A., Merzari, E., and Ferencz, R. M. "SHARP Assembly-Scale Multiphysics Demonstration Simulations." ANL/NE-13/9.

2013 **Jain, R.** and Tautges, T. J. "MeshKit." ANL/MCS-TM-336.

2012 Tautges, T. J., Fischer, P., Grindeanu, I., **Jain, R.**, Mahadevan, V., Obabko, A., Smith, M. A., Hamilton, S., Clarno, K., and Baird, M. "A Coupled Thermal / Hydraulics – Neutronics – Fuel Performance Analysis of an SFR Fuel Assembly." Report to the U.S. Department of Energy.

Software and Open Source

UXarray Lead developer; Python library for unstructured climate grid analysis used by NCAR and DOE labs. GitHub ů Docs.

UXarray MCP MCP server exposing UXarray as typed AI-agent tools with Globus Compute HPC backend; structured provenance on every call. Presented at SciFM26.

FLASH-X I/O and compression lead for a million-line multiphysics simulation engine (astrophysics, combustion, fluid dynamics). R&D 100 Award, 2022.

CANDLE / IMPROVE HPO workflow (Swift/T, mlrMBO, DEAP, Hyperopt) and `improvelib` benchmark framework for cancer drug response; GitHub Actions CI/CD and UNO dual-branch model. R&D 100 Award, 2023.

SFNO PyTorch climate emulator ported to Aurora's Intel XPU stack (60,000+ GPUs): PMIX/PALS launcher, XPU/CUDA AMP, first stable DDP baseline, measured one-node 12-rank run.

MeshKit Principal investigator; open-source C++ reactor mesh generation toolkit. Parallel CoreGen: 101M-element MONJU mesh on 712 processors. Source.

LLM Bench `local-llm-bench-m1max`: reproducible harness and open data for local LLM inference. GGUF bytes-per-token parsing, measured streaming-bandwidth denominator, roofline efficiency across four model architectures. GitHub.

Professional Service

- Program committee member, 1st Workshop on Agentic AI for Large-scale Science (AGENT4SC) at IEEE eScience 2026, Naples, Italy.

- SBIR / STTR proposal reviewer for the U.S. Department of Energy.
- Panelist, “Revolutionizing Public Infrastructure: The Impact of AI and Machine Learning,” 5th Infraday Midwest Event.
- Reviewer for the *Journal of Open Research Software* and NumGrid.
- Program committee member, NumGrid 2020.
- Session chair, Computational Geometries session, MC2015.
- Volunteer, South Side Science Festival, University of Chicago (2023, 2024).

Professional Memberships

- Association for Computing Machinery (ACM), since 2018.
- American Meteorological Society (AMS).
- American Nuclear Society (ANS), 2011 – 2015.

Mentorship

- Mentored students, research aides, and staff on scientific Python, HPC workflows, systems engineering, and reproducible software development.
- Rylie Weaver (Research Aide, 2022 – 2024): IMPROVE project, HPO techniques for cancer drug response prediction.
- Aaron Zedwick (Student, 2023 – 2024): UXarray development, dual mesh routines, and remapping functionality.
- Mark Bartoszek (Systems Administration, 2023): mentoring on systems administration practices at Argonne.
- Brett Rhodes (SULI Summer Student, 2013): MeshKit documentation and examples.
- Dayan Abdulla (Student, 2024 – present): UXarray MCP server development, AI-agent dataset exploration, and Globus Compute integration for HPC execution.